

RYAN THOMAS

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Highly skilled Software Engineer with 6+ years of full-stack engineering experience architecting, scaling, and deploying high-performance applications across aerospace, defense, and cloud platforms. Proven track record of combining enterprise web development with low-level systems engineering—specializing in React frontends paired with C# .NET Core, Go microservices, and Java REST APIs, multi-threaded C/C++ simulation, and distributed real-time messaging. Demonstrated success optimizing SQL database query search times by 4+ minutes at Northrop Grumman, building event-driven streaming pipelines with Azure Event Hubs (Kafka API) and SignalR WebSockets, and orchestrating resilient Generative AI Agents (Gemini LLM) with automated exponential backoff retries and Sentry telemetry. Proficient in cloud infrastructure, Docker containerization, and automated CI/CD pipelines via GitHub Actions.

TECHNICAL SKILLS

Languages & Frameworks: C#, .NET/.NET Core, Go (Golang), C++, Java, React, Python, JavaScript (ES6+), SQL Server, Spring Boot, Qt

Cloud & DevOps: Azure (Container Apps, Cosmos DB, Event Hubs, Blob Storage), AWS, Docker, GitHub Actions CI/CD, Jenkins, Bamboo

Messaging & AI Tools: Apache Kafka, RabbitMQ, SignalR WebSockets, Gemini AI SDK, Sentry Telemetry, Web Audio API

OS & Platforms: Red Hat Enterprise Linux (RHEL 9), VxWorks RTOS, Linux, Git, PowerShell, Visual Studio, TCP/IP

PROFESSIONAL EXPERIENCE

ErreGeTe YGO — Lead Full-Stack Developer

Orlando, FL | 2026 – Present (erregeteygo.com)

- **Full-Stack Architecture & Cloud Infrastructure:** Architected and deployed a multi-service web application built with a React frontend and an asynchronous .NET Core C# API backend, containerized using Docker and deployed to Azure Container Apps via automated GitHub Actions CI/CD pipelines.
- **Go (Golang) Microservice & Concurrency:** Engineered a high-speed Go background microservice utilizing parallel goroutines to offload streaming YDK web scraping and Azure Blob image synchronization logic from the core C# API.
- **Event-Driven Pipeline & Real-Time WebSockets:** Engineered an asynchronous data streaming pipeline leveraging Azure Event Hubs (Apache Kafka API) to decouple data ingestion from core domain logic, broadcasting live community deck creations and activity feeds via SignalR WebSockets.
- **Generative AI Agents & Hand Simulator:** Built interactive AI agents using Gemini LLM to power a Combo Strategy Playbook & Opening Hand Simulator (analyzing 5-card draws, combo branches, and Nibiru/chokepoint vulnerabilities) and an automated Deck Profile Copywriter generating structured Markdown articles.
- **Resilient AI Proxy & Telemetry:** Architected a resilient .NET C# Gemini proxy featuring 3-stage exponential backoff retries for rate limits (HTTP 429), safety category overrides for TCG nomenclature, and Sentry telemetry logging with custom error tags.
- **Database, CDN & Market Valuation:** Migrated local MongoDB stores to an Azure Cosmos DB vCore cluster, configured an Azure Blob Storage CDN with image fallback handling, and integrated a live TCGPlayer/Cardmarket API pricing engine.

Northrop Grumman — Software Engineer

Orlando, FL | Sep 2023 – Present

- **Aircraft Systems Development:** Lead development and design of E-2D Advanced Hawkeye aircraft PCMS and WRMS software systems on Red Hat Enterprise Linux 9 (RHEL 9).
- **Database Optimization:** Reduced query search time of wireless signal data in SQL Server by 4 minutes by consolidating multi-table join logic.
- **Full-Stack Signal Application (React/Java):** Developed a full-stack solution using React and Java to support the development and real-time signal processing of the WRMS application.
- **Aircraft Display Engineering (Qt/C++):** Built data reduction displays on aircraft hardware architecture using the C++ Qt framework to visualize critical flight diagnostics.
- **Scalable Architecture:** Designed scalable, readable, object-oriented code in .NET Core C#, C++, and Java, incorporating multi-threading and unit tests within each subsystem to ensure software efficiency.
- **Signal Telemetry Streaming:** Incorporated Apache Kafka and RabbitMQ for signal topic partitioning, enabling efficient organization and streaming of telemetry messages.
- **DevOps & CI/CD:** Utilized Git and Bamboo for source control alongside Jenkins for automated testing of microservice build scripts, managing ticket resolutions within an Agile/Atlassian framework.

Coleman Aerospace — Software Engineer

Orlando, FL | Feb 2019 – Aug 2023

- **Embedded Simulation:** Built, compiled, and deployed C/C++ simulation code using VxWorks RTOS and Red Hat Linux OS for launch vehicles, satellites, and missile defense power systems.
- **Avionics Documentation & Design:** Developed technical specifications and design documentation associated with avionics test equipment software while enforcing company object-oriented C++ coding standards.
- **Security & Systems Compliance:** Implemented Red Hat Linux updates to align with lab support equipment hardware and updated unclassified/CUI documentation.

EDUCATION

Florida State University — Bachelor of Science in Computer Engineering

Tallahassee, FL